

Building e commerce system

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E-commerce Website / Software

An e-commerce website or software is an **online platform that enables buying and selling of products/services** over the internet.

It handles everything from product display to order fulfillment, payments, taxation, shipping, and customer support.

Think of it as the **digital version of a supermarket** like Bhatbhateni, but running 24/7 on the internet.

Main Functions of E-commerce Software

1. Manage Customers and Orders

- Keeps record of customers, tracks their orders, and processes payments.
- Example: Raj orders a laptop from TechStore. The system saves his details, confirms his order, and sends him a receipt.

2. Manage Products and Catalog

- Stores details of products (name, price, stock, images, description).
- Example: TechStore uploads laptop models, their prices, specs, and availability.

3. Shopping Cart & Transaction Processing

- Allows customers to add items to a virtual cart and checkout.
- Example: Raj adds a laptop + mouse to his cart, checks total price, and pays online.

4. Automates Shipping and Taxation

- Automatically calculates shipping charges, delivery dates, and applicable taxes.
- Example: TechStore system calculates 13% VAT and delivery charge for Raj before payment.

5. Enhances Customer Experience

- Gives personalized suggestions, order tracking, return options, and responsive support.
- Example: Raj gets suggestions for a laptop bag after buying a laptop, plus he can track his package online.

Essential Features of a Good E-commerce Website/Software

1. User-Friendly Design – Simple navigation and clean layout so customers don't get confused.

- Example: Raj easily finds laptops by searching "Dell Laptop" in TechStore.

2. Mobile-Friendly & Responsive – Works smoothly on smartphones, tablets, or desktops.

- Example: Raj can browse TechStore on his phone during college break.

3. Product Comparison – Lets users compare features, prices, and reviews.

- Example: Raj compares Dell and HP laptops before purchase.

4. High-Resolution Media – Clear product photos and videos.

- Example: TechStore shows multiple angles of the laptop with a demo video.

5. Special Offers and Discounts – Coupons, seasonal sales, or free shipping.

- Example: Raj gets "Dashain Offer: 10% off + free mouse."

6. Strong Security – Protects personal and payment data using SSL/TLS.

- Example: Raj's credit card details are encrypted while paying online.

7. Advanced Payment System – Multiple payment options like cards, wallets (eSewa, Khalti), COD.

- Example: Raj pays via eSewa, while another customer pays Cash on Delivery.

8. Clear Return & Refund Policy – Builds trust in case of damaged or wrong products.

- Example: Raj knows he can return the laptop within 7 days if it's faulty.

Building Catalogs in E-commerce

A **product catalog** in e-commerce is an **organized list of products with details like name, description, price, availability, and images/videos**. It acts as the **digital shelf** of an online store, where customers browse and decide what to buy.

A **good product catalog** is the **key to success** in e-commerce because it influences customer trust, shopping experience, and conversion rates.

Importance of Product Catalog

- Provides complete and clear product information.
- Helps customers compare products easily.
- Enhances customer experience with photos, videos, and specifications.
- Reduces confusion and customer complaints.
- Makes updating stock, discounts, and product details easier for the business.

Example: Raj visits **TechStore's** website. He sees laptops with images, detailed specifications, prices, warranty info, and stock availability. This helps him decide quickly.

Types of Catalogs

1. Static Catalog

- A **simple HTML-based catalog**.
- Product list and details are written directly into web pages.
- To **add, edit, or delete** items, the company must manually edit the HTML code.
- Not flexible for large businesses.

Features:

- Single or related web pages with product details.
- Each item has a photo and short description.
- Best suited for **small businesses with few products**.

Example (Static Catalog):

TechStore is a small startup selling 5 laptop models. The owner creates an HTML page showing:

- Laptop Name: Dell Inspiron
- Price: Rs. 65,000
- Short Description + 1 photo

If Raj buys one model and stock finishes, the owner must **manually edit the HTML file** to update availability.

2. Dynamic Catalog

- Product details are stored in a **database** (MySQL, PostgreSQL, etc.).
- Website fetches information from the database whenever a user browses.
- **Easier to manage:** merchants update items from an **admin dashboard** without editing HTML.
- Suitable for medium to large businesses with hundreds/thousands of products.

Features:

- Can display multiple photos, specs, customer reviews, related products.
- Auto-update when stock changes.
- Can show personalized recommendations.

Example (Dynamic Catalog):

TechStore grows and now sells 500+ products (laptops, mobiles, accessories).

- Raj searches "Dell Laptop" → the database instantly filters all Dell models.
- If 20 laptops go out of stock, the system automatically hides them or marks them "Out of Stock."
- Admin can update prices or add discounts through the dashboard

Building Shopping Cart in E-commerce

An **online shopping cart** is a virtual basket on an e-commerce website that allows customers to

collect, review, and manage products before purchasing.

It acts as a **bridge between product catalog and transaction processing**.

Features of a Shopping Cart

1. **Add to Cart** → Customer clicks “Add to Cart” when selecting a product.
Example: A user adds an iPhone 15 from Daraz into the cart.
2. **Checkout Button** → Provides an option to proceed to payment and delivery.
Example: Customer clicks “Checkout” to confirm the iPhone 15 purchase.
3. **Continue Shopping** → Lets users return to the catalog to add more items.
Example: After adding iPhone 15, customer goes back to add AirPods.
4. **Remove Products** → Customers can delete unwanted items.
Example: Customer removes AirPods before final payment.
5. **Edit Product** → Update quantity, size, or color.
Example: Customer changes iPhone 15 color from Black to Blue.
6. **Search Products in Cart** → Easy navigation when cart has multiple items.
7. **Edit Order Details** → Change delivery location, gift wrap, etc.

Transaction Processing

Once the shopping cart is finalized, the next step is **transaction processing**.

Steps in Transaction Processing

1. **Cart Verification**
→ Customer reviews all products, price, tax, and shipping.
Example: Daraz shows total = iPhone 15 (Rs. 150,000) + Rs. 200 delivery charge.
2. **Payment Method Selection**
→ Options: Debit/Credit Card, eSewa, Khalti, Cash on Delivery.
3. **Payment Authorization**
→ If using card: Secure protocols like SSL/TLS or SET verify details.
Example: Customer enters NIC Asia Bank credit card → Bank verifies balance.
4. **Funds Transfer**
→ Debit from customer’s account, credit to Daraz’s account.
5. **Order Confirmation**
→ Customer gets an order receipt & Daraz prepares delivery.

Security in Transaction Processing

- **SSL/TLS encryption:** Protects sensitive data during transmission.
- **SET Protocol (Secure Electronic Transaction):** Provides cardholder authentication & digital certificates.

Development of E-commerce Website/Software

Before building an e-commerce site, firms need **planning** (budget + timeline) and a clear **vision** (problem statement + development method).

Factors to Consider

- **Management** – who runs daily operations.
- **Software** – platform (Shopify, Magento, or custom).
- **Hardware** – hosting/cloud servers.
- **Design** – mobile-friendly, secure.
- **Telecom** – internet, payment gateway links.
- **Human Resources** – developers, testers, support staff.

Example: Daraz uses AWS servers, a custom app, and local couriers.

Steps (SDLC Approach)

1. **System Analysis / Planning** – requirements, budget, tech.
Daraz: COD, card payments, discounts.
2. **System Design** – UI/UX, database, payment flow.
Daraz: filters, cart, secure login.
3. **Building** – coding, database setup, API integration.
Daraz: stock system + eSewa/Khalti integration.
4. **Testing** – check speed, security, order flow.
Daraz: iPhone 15 order deducts correctly.
5. **Implementation** – launch + monitor.
Daraz: added “DarazMall” verified stores.

Integration of E-commerce Software with Other Systems

E-commerce software rarely works alone — it needs to **integrate with databases, applications, ERP systems, and payment gateways** to run smoothly.

1. Database Integration

- A **database** stores structured information like product details (size, color, price).
- Small businesses → **MS Access**.
- Large businesses → **IBM DB2, SQL Server, Oracle**, or distributed databases.
- **Example (Daraz Nepal)**: When you search “Red Nike Shoes, size 42”, the database quickly fetches all matching products.

2. Application Programs Integration

- **Application Programs** = software that performs specific functions (invoice generation, payroll, payment processing).
- **Application Server**: Receives requests from the web server and applies **business logic** (rules of the business).
- Integration = linking different apps so they share data automatically.
- **Example (Daraz)**: When a customer checks out, the system calls:
 - Payment app → deduct money.
 - Inventory app → reduce stock.
 - Invoice app → email receipt.

3. Integration with ERP Systems

- **ERP (Enterprise Resource Planning)** = software that integrates **accounting, logistics, manufacturing, marketing, planning, etc.**
- Big vendors → **Oracle, SAP** (expensive).
- Smaller firms → use ERP on **subscription/SaaS model** (pay monthly, use via web browser).
- **Example**: A Nepali clothing brand using SaaS ERP can track fabric purchases, online orders, delivery schedules, and accounts in one system — instead of using 4 separate apps.

4. Integration with Payment Gateways

- A **Payment Gateway** authorizes & processes online payments.
- Ensures secure transaction flow between **customer ↔ merchant ↔ bank**.
- Uses encryption + security protocols (SSL/TLS).
- 4 methods of integration differ based on:
 - **PCI DSS compliance** (whether the business must store/handle card data).
 - **User experience** (smooth checkout vs. redirects).
- **Example (Nepal)**: Daraz integrates with **eSewa, Khalti, Visa/MasterCard, FonePay** → customers can pay securely during checkout.

Using Open Source CMS for E-commerce Applications

Content Management Systems (CMS) help businesses manage the large amounts of text, images, videos, and other media files needed for running an online store. Previously, all information—like reports, schedules, analyses, and memos—was stored on paper. Now, CMS allows companies to organize, control, and move this information online efficiently.

Benefits of CMS in E-commerce:

- Organizes and updates product information, images, and descriptions.
- Controls who can add, edit, or delete content.
- Publishes content directly to the website or social media platforms.
- Makes managing online operations easier and faster.

Open Source CMS:

- The source code is publicly available, so anyone can use, modify, and enhance it.
- Examples include **WordPress with WooCommerce, Magento Open Source, and Joomla.**
- Some CMS require basic development skills to customize themes, features, or plugins.

Example:

Imagine a Nepalese clothing store selling online through its website:

- Using **WordPress + WooCommerce**, the store can upload product images, add prices, manage inventory, and create promotional banners.
- When the store wants to change a product description or launch a new offer, the CMS allows them to update it instantly without coding everything from scratch.